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1 Battery pathway (deintercalation electrode characteristics)

A bulk pathway that runs after the gen/csp + phase-diagram stages, as an alternative to the catalysis (surface builder + adsorbates) branch. It takes the stable host structure(s) containing a working ion A (Li, Na, K, Mg, Zn), removes A stepwise, and computes electrode characteristics from the energies $E(x)$ of $A_x \cdot \text{Host}$ on a pseudo-binary convex hull.

Submission reuses the existing schema with **no DB migration**:

```
{"composition": "LiFePO4", "reaction": "battery", "reaction_path": "Li"}
```

reaction_path carries the working ion, and the per-ion step status nests in DBComposition.step_status["battery"] exactly like the per-(reaction, pathway) adsorbates status. Surface builder, adsorbates and the catalysis calculators are skipped for battery submissions (bulk pathway, no slabs).

1.1 What tier 1 computes (MLIP-only, no new calculation plugins)

For each host and working ion:

- **Voltage profile** from the pseudo-binary convex hull over x . Between adjacent hull vertices $x_1 \rightarrow x_2$:
$$V = -[E(x_2) - E(x_1) - (x_2 - x_1) \cdot \mu_A] / (z \cdot (x_2 - x_1) \cdot e)$$
with μ_A = energy/atom of the elemental ion metal (same MLIP, via the existing elemental-reference machinery) and $z = 1$ (Li/Na/K) or 2 (Mg/Zn).
- **Average voltage** over the full range (depends only on the end members).
- **Gravimetric capacity** $Q = n \cdot z \cdot F / (3.6 \cdot M)$ in mAh/g, M = molar mass of the discharged formula unit (standard cathode convention: LiFePO₄ $\rightarrow$ 170 mAh/g).
- **Volumetric capacity** (mAh/cm³) from the discharged-state volume.
- **Specific energy** (Wh/kg) = $Q_{\text{grav}} \times V_{\text{avg}}$.
- **Volume change** between charged and discharged end members (cycling proxy).
- **Charged-endpoint stability**: $e_{\text{above_hull}}$ of the empty host against the chemsys hull the gen path already built. Far above the hull = conversion-like chemistry $\rightarrow$ flagged, not silently reported as intercalation.
- **Framework integrity**: StructureMatcher on the host sublattice (ion removed) charged vs discharged, plus a volume-collapse guard. Detects layer gliding / reconstruction that invalidates the intercalation picture.

All of tier 1 rides on the existing MLIP relax workchains (MACE / UMA / MatterSim, job_type: relax) — zero new AiiDA calculation or parser plugins.

1.2 Method choices locked in

1. **v1 is deintercalation-only.** The submitted formula must contain the working ion (LiCoO₂, NaFePO₄, ...). Insertion into an empty host (TiO₂ + Li) needs interstitial-site search and is a v2 feature.
2. **One common supercell for all x**, built once from the host primitive cell (capped at `supercell_max_atoms`), so every grid point is an exact fraction k/N of the N ion sites — no incommensurate occupancies.
3. **Enumeration is pymatgen-only** (no `enum.x` binary, no `icet` requirement): partial occupancy x on the ion sublattice, orderings ranked by Ewald electrostatics (oxidation states from `BVAnalyzer` / composition guess); if no oxidation states can be assigned (metallic hosts), fall back to symmetry-distinct sampling with a fixed seed. `icet` enumeration can replace this later without touching the workflow.
4. **Fail loudly.** Framework collapse and hull-unstable charged endpoints are recorded as flags on the DB row; a voltage from a structure that fell apart is never reported as a clean result.
5. **PBE-head caveat.** MACE `omat_pbe` is plain PBE: absolute voltages of TM-oxide redox couples sit systematically low (roughly 0.5–1 V). Fine for *ranking*; publishable absolute voltages need the tier-2 DFT (+U) stage.

1.3 Architecture / file map

piece	file	notes
pure calculator	<code>workchains/batt.py</code>	hull, voltages, capacities, volume change; no AiiDA imports , unit-tested standalone (same family as <code>oer.py/co2rr.py</code>)
enumeration helper	<code>workchains/battery_enum.py</code>	supercell choice + per-x orderings; pure pymatgen, unit-tested standalone
workchain	<code>workchains/battery.py</code>	BatteryWorkChain: setup -> enumerate -> bundled MLIP relax fan-out -> analyze -> store
DB table	<code>db/tables.py::DBBatteryPath</code>	one row per (composition, working_ion, host); results in Postgres, not AiiDA storage (ephemeral)
entry point	<code>setup.json</code>	<code>battery =</code>
main gating	<code>workchains/main.py</code>	<code>uvsib.workchains.battery:BatteryWork</code> <code>should_run_battery on</code> <code>reaction == "battery";</code> surface builder + adsorbates skip battery submissions
tests	<code>tests/test_batt.py</code> , <code>tests/test_battery_enum.py</code>	run in any env with pymatgen, no AiiDA / no DB
local smoke	<code>tests/smoke_battery_mace.py</code>	standalone LFP/LCO end-to-end with a local MACE (not collected by pytest)

The relax fan-out bundles the working-ion elemental reference through the same `missing_element_references` / `element_reference_entries` machinery the gen and csp paths use, so the anode reference (Li bcc, Na bcc, ...) is relaxed on the same MLIP — methodologically consistent voltages for free.

1.3.1 input.yaml block (all keys optional; module runs without the block)

```
battery:
  model: MACE                      # MLIP for the config relaxations (defaults to bulk_relax)
  head: Default
  fmax: 0.05
  max_steps: 200
  n_x_steps: 4                    # intermediate compositions between charged and discharged
  max_configs_per_x: 8            # orderings kept per grid point (Ewald-ranked)
  supercell_max_atoms: 128
  max_hosts: 3                    # stable_struct candidates to sweep (<= MAX_NUM_BULK)
```

1.4 Status

- ☒ plan agreed (2026-07-21)
- ☒ workchains/batt.py pure calculator + unit tests
- ☒ workchains/battery_enum.py enumeration helper + unit tests
- ☒ BatteryWorkChain + DB table + entry point + main.py gating
- ☒ local end-to-end smoke (2026-07-21, MACE-MP-0 medium, RX 6900 XT; enumerate -> cell+position relax -> batt.py, ~50 s per material): - LiFePO4 (mp-19017): V_avg 3.438 V (exp 3.45 - MPtrj carries Fe +U), Q 169.9 mAh/g (theoretical, exact), 584 Wh/kg, dV -5.2% (exp ~ -7%), framework intact; the 3.25-3.60 V staircase around the experimental flat plateau is the finite-supercell ordering artifact, as expected - LiCoO2 (mp-22526): V_avg 3.277 V (low vs ~3.9 exp - the documented MLIP voltage caveat for Co redox), Q 273.8 mAh/g (theoretical, exact), 898 Wh/kg, dV +4.6% - correctly POSITIVE (layered c-axis expansion on delithiation, opposite sign to olivine), O3 framework intact - determinism confirmed: repeated runs bit-identical energies
- ☐ deploy: python -m uvsib.db.tables to create db_battery_path, reinstall for the new entry point, submit a battery entry
- ☐ tier 2: DFT (+U) verification stage on hull-vertex configurations only (analog of pd_verification; battery_fw recipe)
- ☐ tier 2: MLIP-NEB migration barriers (dilute vacancy / ion hop) — **shared infrastructure with the planned catalysis NEB**, design them together
- ☐ v2: insertion mode for ion-free hosts (interstitial site search)

1.5 Tier-2 notes (for when we get there)

- **NEB**: one generic MLIP-NEB workchain (images from IDPP interpolation, climbing image, existing relax calcjobs extended with a job_type: neb) serves both battery migration barriers and catalysis reaction barriers — do not build two.
- **DFT verification**: statics/short relaxes on the hull vertices only (typically 3-6 structures per host), PBE+U per the battery_fw settings; MP2020-style corrections only matter if we mix with MP entries — internal voltages need consistent U, nothing else.
- **Electrolyte window / interface reactivity** (grand-potential hulls): out of scope until someone asks.